DE–3032

Sub. Code
13

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2017.

DIGITAL COMPUTER FUNDAMENTALS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Convert octal Number $(526.06)_8$ to Hexadecimal.

2. State and Prove Demorgan's theorem.

3. Draw the circuit diagram of decoder. Explain.

4. Explain the combinational logic with example.

5. Discuss the Full Adder circuit diagram with truth table.

6. Explain the operation of JK flip-flop.

7. Write short note on Gated flip-flops.

8. Briefly explain about any three output devices.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain about various number systems with suitable examples.

10. Simplify the following expression in Product of Sum.

$F(A, B, C, D) = \Pi (0, 1, 2, 3, 4, 10, 11)$

11. Explain the operation of de-multiplexer with neat diagram.

12. Discuss about various types of shift register with a diagram.

13. Describe about BCD counter with neat sketch.

14. Explain the classification of computers.

15. Write short note on:

    (a) Random Access Memory (RAM)

    (b) Read Only Memory (ROM).

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Sub. Code
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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Year Integrated) DEGREE EXAMINATION, DECEMBER 2017.

C AND DATA STRUCTURES

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain about various data types in C.
2. Discuss in detail about the multi dimensional arrays in C. Give example.
3. Explain the concept of 'structure' within the 'structure'. Give example.
4. Write a program to find the factorial of N using recursion.
5. Describe the concept of passing pointers to function.
6. Explain the representation of Queue. What are its applications?
7. Discuss about the singly linked list with example.
8. Define Tree. What are the applications of Tree?

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain about 'for' and 'while' and 'do-while' statements with suitable example.

10. (a) Distinguish between 'break' and 'continue' statements. (7)

    (b) Explain about 'switch' statement. Give example. (8)

11. (a) Discuss about various library functions in C. (8)

    (b) Write a program to check the given string is palindrome or not. (7)

12. Write short note on:

    (a) Dynamic Memory Allocation (7)

    (b) Opening and closing a file. (8)

13. Define Stack. Explain the algorithm to perform the infix to postfix conversion.

14. What is Binary tree? Explain the linked representation of Binary tree. Give examples.

15. Define Circular Queue. Explain the algorithm to perform the operations of Circular Queue.

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DE-3034

Sub. Code
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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2017.

DISCRETE MATHEMATICS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Construct the truth table for $(P \vee Q) \vee \rceil P$.

2. Explain the concept of Tautology with an example.

3. If $A_1 = \{\{1, 2\}, \{3\}\}$, $A_2 = \{\{1\}, \{2, 3\}\}$ and $A_3 = \{\{1, 2, 3\}\}$ then show that A_1, A_2 and A_3 are mutually disjoint.

4. Write the members of $\{a, b\} \times \{1, 2, 3\}$.

5. Show that the functions $f(x) = x^3$ and $g(x) = x^{\frac{1}{3}}$ for $x \in R$ are in verses of one another (R is a set of real numbers).

6. Let $f(x) = x + 2$, $g(x) = x - 2$ and $h(x) = 3x$ for $x \in R$, where R is a set of real numbers. Find $g \circ f$, $f \circ g$, $f \circ f$ and $g \circ h$.

7. Show that every finite semigroup has an idempotent.

8. How graph coloring is done?

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Show that the following are equivalent formulas

(a) $P \vee (P \wedge Q) \Leftrightarrow P$

(b) $P \vee (\neg P \wedge Q) \Leftrightarrow P \vee Q$.

10. Show that $S \vee R$ is tautologically implied by $(P \vee Q) \wedge (P \rightarrow R) \wedge (Q \rightarrow S)$.

11. If *A, B* are two sets then show that $A \subseteq B \Leftrightarrow A \wedge B = A$.

12. Given $S = \{1, 2, 3, 4\}$ and a relation R on S defined by $R = \{\langle 1, 2\rangle, \langle 4, 3\rangle, \langle 2, 2\rangle, \langle 2, 1\rangle, \langle 3, 1\rangle\}$ show that R is not transitive.

13. Show that the function $f(x) = \begin{cases} x/2 & when\ x\ is\ even \\ (x-1)/2 & when\ x\ is\ odd \end{cases}$ is primitive recursive.

14. Show that the set of all elements a of a group $\langle G, *\rangle$ such that $a * x = x * a$ for every $x \in G$ is a subgroup of *G*.

15. How to represent the graph in computer memory?

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DE–9856

Sub. Code
16

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2017.

C PROGRAMMING — LAB

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by LOT system.

1. (a) Write a C program to convert a given number into words for numbers 1 to 5

   Example 1 to ONE

   2 to TWO

   5 to FIVE

   (b) Write a C program to read 10 values to an array variable. Use pointers to locate and display each value.

---------- Cut here ----------

2. (a) Write a C program to find the roots of a quadratic equation $ax^2 + bx + c = 0$.

   (b) Write a C program to find a word is PALINDROME or not.

3. (a) Write a C program to find the roots of a quadratic equation $ax^2 + bx + c = 0$ using function definition.

(b) Write a C program to find the factorial of a given number using recursion.

---------- Cut here ----------

4. (a) Write a C program for the following matrix manipulations

(i) Addition

(ii) Multiplication using function.

(b) There are 10 students in a class. Their names and marks in three different subjects are given. If a student takes more than 40 marks in each subject, then he is declared to "PASS" otherwise "FAIL". Write a C program to do above using structure concept.

---------- Cut here ----------

5. (a) A man is paid at the hourly rate of Rs. 15 per hour for the first 45 hours worked. Thereafter, overtime is paid at 1.5 times the hourly rate for the next 25 hours and 2 times the hourly rate for further hours worked. Write a C program to input the number of hours worked per week. Calculate and print his gross weekly range.

(b) Write a C program to implement any five string operations using switch statement.

DE–9857

Sub. Code
17

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Year Integrated) DEGREE EXAMINATION, DECEMBER 2017.

DATA STRUCTURES LAB

Time : Three hours Maximum : 100 marks

ONE question should be given to each candidate by LOT system. Each sub-division carries 50 marks.

1. (a) Write a program to implement push and pop operations an stack.

   (b) Write a program to sort the given set of elements using insertion sort.

---------- Cut here ----------

2. (a) Write a program to implement insert and delete operations on linked list structure.

   (b) Write a program to sort the given set of elements using selection sort.

---------- Cut here ----------

3. (a) Write a program to implement insert and delete operations on queue using array concept.

   (b) Write a C program to sort 10 numbers in ascending order with naming of variable and the value before and after sorting.

4. (a) Write a C program to evaluate the given mathematical expression using stack.

   (b) Write a program in C to search the given element in an array using binary search.

---------- Cut here ----------

5. (a) Write a C program for linked list implementation of queue operations.

   (b) Write a C program to sort the given set of elements using insertion sort.

---------- Cut here ----------

6. (a) Write a C program to implement insert and delete operations on stack.

   (b) Write a C program to copy the contents of one file to another file.

---------- Cut here ----------

7. (a) Write a C program to implement insert and delete operations on queue using array concept.

   (b) Write a menu driven program in C to search an element using linear and binary search methods.

---------- Cut here ----------

8. (a) Write a C program to implement insert and delete operations on linked list structure.

   (b) Write a program to copy the contents of one file to another file.

DE–3035

Sub. Code
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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Year Integrated) DEGREE EXAMINATION, DECEMBER 2017.

OBJECT ORIENTED PROGRAMMING AND C++

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. Mention the basic concepts of object oriented programming.

2. Explain the usage of cin and cout streams.

3. What is meant by data encapsulation?

4. How to use constructors?

5. List the problems with pointer reference.

6. Give the purpose of dynamic constructor.

7. Explain the concept of polymorphism in compile time.

8. Discuss on static and dynamic binding.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following questions.

9. Discuss the usage of manipulators and functions in C++.

10. How to define and declare the member functions?

11. Explain the usage of copy constructor.

12. Mention the pitfalls of operator overloading.

13. What is inheritance? Discuss its types with examples.

14. Write a program in C++ to implement any three string functions.

15. Write a C++ program to implement the concept of overloading binary operators.

DE–3036

Sub. Code
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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2017.

OPERATING SYSTEMS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. Mention the functionality of an operating system.

2. Explain the concept of system calls.

3. Define (a) Process (b) Processes (c) processor.

4. Discuss on Round Robin scheduling.

5. List the goals of I/O software.

6. How to detect and recover the deadlocks?

7. Discuss the concept of multiprogramming with fixed partitions.

8. Discuss on the performance of file system.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following.

9. Narrate the history of operating systems.

10. Describe the structure of the operating system.

11. Explain the concept of Race condition.

12. Discuss the following
    (a) Policy driven scheduling.
    (b) Two level scheduling.

13. Sketch the algorithm for page replacement.

14. Discuss the analysis of swapping systems.

15. How to provide protection mechanisms for File Systems?

DE–3037

Sub. Code
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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2017.

RELATIONAL DATABASE MANAGEMENT SYSTEM

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. Explain briefly about the Schema and Subschema in DBMS.

2. Explain the two kinds of Data Independence.

3. Write the Merits and Demerits of using relational databases.

4. Write short notes on concept of relation model.

5. Mention some benefits of using Oracle as database.

6. Explain the important factor while evaluating your system's level of CPU utilization.

7. Explain briefly the data manipulation language statements.

8. How to create database links to access data and schema objects in a distributed database system?

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain with neat illustration about the three level architecture of DBMS.

10. Discuss briefly about the relational database model with suitable example.

11. Discuss in detail about the data control facilities in DBMS with example.

12. Give the details of the following keys :

    (a) Super key

    (b) Candidate key

    (c) Primary key

    (d) Foreign key.

13. Explain briefly about joining multiple tables in a query.

14. Write short notes on the following :

    (a) Modifying tables

    (b) Renaming tables.

    (c) Copying another table.

    (d) Dropping a table.

15. Discuss in detail about Embedded SQL commands and its uses.

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DE–9858

Sub. Code
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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2017.

C++ – LAB

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

1. (a) Create a class employee that contains a employee number, name and address. Write a program in C++ to get the 'n' number of employee details and display all employee details in namewise sorted order.

   (b) Write a program using polymorphism to calculate the square of any two numbers of type int, float, double and long.

---------- Cut here ----------

2. (a) Write a program to process students marks with the help of classes. The class has private variables, for name, mark 1, mark 2 and mark 3. It has two member functions – getdata( ) to get input, result( ) to print the result. All subjects marks must be greater than or equal to 50 for pass, otherwise fail.

   (b) Create a four function (addition, subtraction, multiplication and division) calculator using C++.

3. (a) Write a C++ program to find the different and total length of given two various tubes specifies in meters and centimeters using operator overloading concept.

(b) Imagine a publishing company that markets both books and audio-cassette versions of its works. Create a class publication that stores the title and price of a publication. From the class derive two classes-book, which adds a page count and tape, which adds a length count. Each classes have a getdata( ) function to get its data from the keyboard and a putdata( ) function to display its data. Write a main( ) program to test the book and tape classes.

---------- Cut here ----------

4. (a) Write a C++ program to concatenate two given strings using dynamic constructors.

(b) Write a menu driven C++ program to add and subtract two given matrices using operator overloading.

---------- Cut here ----------

5. (a) Write a C++ program to print the multiplication table of a given number.

(b) Create a class called TIME that has integer data elements for hours, minutes and seconds. The constructor should initialize these elements to specified value, if given, otherwise to 0. A member function should display it in 10 : 45 : 40 format.

6. (a) Write a function called power( ) that raising a number n to power p. Use default argument of 2 for p, so that if this argument is omitted. Write a main( ) function that gets values from the user to test this function.

(b) Write a program to perform complex number arithmetic operations using operator overloading.

DE–9859

Sub. Code
27

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2017.

RELATIONAL DATABASE MANAGEMENT SYSTEMS — LAB

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

1. (a) A hospital maintains blood donors record information in a file. The file contains the following fields.

Donor number, name, age, address, place of birth, blood group and mobile number.

Write a program to print out the number, name address and mobile number of the donors for the following categories.

(i) Blood donors having blood group A+

(ii) Blood donors having blood group O+ and the age between 20 and 25

(iii) Female donors with B blood group in the age between 18 and 25.

(b) Write a program to calculate the commission for the given amount.

if the amount > 50000; commission is 20%

if the amount > 25000 and $\leq$ 50000; commission is 10%

Otherwise commission is 5%

The basic table is com and the corresponding fields are, accno, amount and commission.

2. (a) A salary statement contains.

Name, Basic pay, Allowances, Total deduction, Gross pay and Net pay.

DA = 72% of Basic pay

Gross pay = Basic pay + DA

Deductions = PF + IT + LOAN

IT is calculated on the basis of annual income

Annual salary	IT
upto 2.5 lakhs	Nil
2.5 laks to 5 lakhs	5%
5 laks to 10 lakhs	20%
Above 10 lakhs	30%

Write a program in SQL to prepare salary report for 5 employees.

(b) The base table is having the columns of Regno, name, marks and rank. Enter Regno, name and marks in the base table upto 10 records. Write a program to update the base table while allocating rank. Assume your own classifications for assigning rank to the student.

3. (a) Write a program for hospital billing system with the following fields :

Patient number, name age, patient-type (IN/OUT), consulting charge, X-ray charge, blood test charge, other test charges and total fee.

Prepare reports with the following conditions :

(i) The patients undergone for blood test

(ii) The patients who take X-rays

(iii) The patients who belong to In patient category.

The common fields to be included in the above reports : Patient number, name, age, charges and total charges.

(b) Write a program to evaluate the HRA, IT, Gross pay and Net pay.

Base table is PAY and the corresponding fields are employee number, name, basic pay, DA, HRA, IT, Gross pay and Net pay.

Input from the base table is employee number, name, basic pay.

4. (a) An airline reservation database contains the reservation table and personal table.

Reservation table : Flight No., class, passenger name and Seat No.

Personal table : Passenger name, sex, age, marital status and nationality.

Write a program to prepare the following reports.

(i) List the passenger names with seat number according to class and flight numbers.

(ii) Total number of married female passengers in a particular flight.

(iii) List out the male passengers in the age between 40 and 60 for all flights.

(b) Write a program to process the Xth class results for the following conditions :

English, Tamil, Science, Maths, Social > = 40 pass

Total is > = 300; I class

> = 250 and < 300; II class

otherwise III class

The base table is MARK and the fields are :

Regno, name, English, Tamil, Maths, Science, Social, TOTAL, RESULT.

---------- Cut here ----------

5. (a) Write a program to prepare a mark statement of a student using student table and mark table. Assume your own data.

(b) Write a program to calculate electrical charges as per the rates given below :

Industry Rs. 2.00/unit

Agriculture Rs. 1.00/unit

Domestic Rs. 1.50/unit

Assume your own fields for base table.

DE–3038

Sub. Code
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DISTANCE EDUCATION

M.Sc. (5 Years Integrated) (Software Engineering) DEGREE EXAMINATION, DECEMBER 2017.

COMPUTER NETWORKS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Describe the uses of computer networks.

2. Write about ISDN.

3. Describe any four framing methods.

4. Describe the bit-map protocol and the problem associated with the protocol.

5. Compare virtual circuit with datagram subnets.

6. Describe the address resolution protocol.

7. How can you establish a connection using a three-way handshake?

8. Describe the different categories of encryption methods.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the different guided media that are used for transmission.

10. With suitable examples, explain error-correcting and error-detecting codes.

11. Describe the Dijkstra's algorithm to compute the shortest path through a graph.

12. Write about transmission control protocol in detail.

13. What are the issues related to network performance? Explain.

14. Give an overview of what e-mail systems can do and how they are organized.

15. Discuss about MPEG standard.

DE–3039

Sub. Code
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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2017.

INTERNET AND JAVA PROGRAMMING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Briefly explain about various search engines and web browsers.

2. How to define and use java packages? Explain with example.

3. Explain about various layout managers in AWT. Draw diagram.

4. Design HTML code for user registration form.

5. Write a JavaScript code to display month names using arrays.

6. Explain about input stream classes with example.

7. Write short note on: Remote Method Invocation (RMI).

8. Define Applet? Write an applet program to accept username and password.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain in detail about the object oriented programming concepts.

10. (a) Define Servlet. Discuss the life cycle of the Servlet. (7)

(b) Explain about various AWT controls. (8)

11. Write short note on:

(a) Domain Naming System (DNS) (5)

(b) File Transfer Protocol (FTP) (5)

(c) Internet Relay Chat (IRC) (5)

12. Discuss about various control Structures. Illustrate using javascript.

13. Explain about various file stream classes in java. Give example.

14. Explain in detail about exception handling mechanism in java.

15. (a) Describe the steps involved in JDBC database connectivity? Explain. (10)

(b) What the characteristics of java beans? Explain. (5)

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DE–3040

Sub. Code
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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Year Integrated) DEGREE EXAMINATION, DECEMBER 2017.

SOFTWARE ENGINEERING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. Discuss the role of Management in Software development.

2. What are the uses of Process Technology Tools?

3. Give a brief note on Software Quality Assurance plan.

4. What are the Project Planning activities? Explain.

5. How do you configure a software project? Discuss.

6. What is meant by software review? Explain.

7. How to represent information flow? Explain with example.

8. List the possible errors of Black Box Testing.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following.

9. Describe the various Phases of Unified Process. (15)

10. Explain the various basic principles of software project scheduling. (15)

11. What are the requirements of Engineering Process Functions? (15)

12. Explain the different elements of Design Model. (15)

13. Discuss the Basic Design principles of Class-Based Components. (15)

14. (a) What errors that occur commonly during the System Testing? (7)

    (b) What problems may be encountered when Top-Down Integration is chosen? (8)

15. Describe the importance of Project metrics. (15)

DE–3041

Sub. Code
34

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2017.

COMPUTER GRAPHICS AND MULTIMEDIA

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. Discuss the various interactive input methods.

2. Explain the Bresenham's line drawing algorithm.

3. Write a note on Spline curves.

4. Describe the properties of the Bezier curves.

5. Discuss the RGB color model.

6. Explain about any two multimedia hardwares.

7. Write a note on authoring tools.

8. Discuss, in detailed, about the media on demand.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following.

9. Describe the Bresenham's Circle drawing algorithm.

10. Discuss the Polygon clipping algorithm.

11. Explain the 2D-Transformation with matrix notation.

12. Write a note on the hidden surface elimination methods.

13. Discuss, in detailed, about various Multimedia hardwares.

14. List the various applications of computer graphics and multimedia.

15. Elucidate the Video Conferencing, and list its various applications.

DE–3042

Sub. Code
35

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2017.

UNIX AND SHELL PROGRAMMING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. List the features of the Unix Operating System. (8)

2. Illustrate the general format of the directory and file operations of any four Unix commands. (8)

3. Explain the filter and pipe concepts in Unix operating system. (8)

4. How to manage file security in Unix operating system? Illustrate. (8)

5. Discuss the general format of different types of the 'grep' command with example. (8)

6. Explain about any four string functions of the Unix Operating System with its usage. (8)

7. Write a script to copy the content of one file to another. (8)

8. Discuss the different types of variable declaration in Unix Operating System. (8)

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following.

9. Describe the Unix file system and file handling utility commands with examples. (15)

10. Explain the architecture of the Unix Operating System. (15)

11. Write a shell program to compare the contents of two files, and count the words in the files. (15)

12. Discuss the various mathematical functions available in Unix with examples. (15)

13. Illustrate the various control structures in shell programming. (15)

14. How to define user-defined functions in Unix? Explain with examples. (15)

15. Explain the different types of system calls for file management in Unix operating system. (15)

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DE–9860

Sub. Code
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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated)
DEGREE EXAMINATION, DECEMBER 2017.

INTERNET AND JAVA – LAB

Time : Three hours Maximum : 100 marks

The Examiner has to choose any ONE question.

Each subdivision carries 50 marks.

1. (a) Write a Java script to create a colour palette and display the text in the colour chosen from the palette with proper background colour.

   (b) Write a VBScript, which displays a Calender for the given Month and Year.

---------- Cut here ----------

2. (a) Write a Swing program to create buttons with :

   (i) Tool tip text

   (ii) Image

   (iii) Border

   (iv) Short cut key.

   (b) Write a VBscript to perform the following :

   (i) Get a password

   (ii) Check the given password

   (iii) Change the existing password.

3. (a) Write a Java script to create a colour palette and display the text in the colour chosen from the palette with proper background colour.

(b) Create a form in HTML containing the following fields and then perform the validation of each field using VBScript.

Name – textbox (validation : Only text)

Address – textbox (validation : number with text)

Date of Birth – a combo box (one for each, day, month and year)

Email – textbox (validation : Only text with symbol)

---------- Cut here ----------

4. (a) Write a Java program using Applet to display a dialogue menu.

(b) Develop a single page advertisement in HTML for a shop to be opened shortly.

---------- Cut here ----------

5. (a) Create a Web page in the format of the front page of a News Paper using Text link in HTML. Align the Text with colours.

(b) Write a Swing program to create a Tabbed panel.

6. (a) Develop a Web page using HTML for a job recruitment agency for software development companies.

(b) Write a Java script to create an Order Form to purchase the House article.

---------- Cut here ----------

7. (a) Write a VBScript to perform the following :

(i) Display the current date.

(ii) Find the difference between the two dates.

(iii) Find age of a person for his date of birth submitted.

Use the date format like DD/MM/YY.

(b) Create a Web page with the message 'M.Sc. SOFTWARE ENGINEERING, ALAGAPPA UNIVERSITY'.

---------- Cut here ----------

8. (a) Design and develop a Web page for a College. The Web page should contains the information related to various Departments in the college and courses offer by them.

(b) Write a Java Script to create a window by using the confirm message.

9. (a) Develop a Web page using HTML for a job recruitment agency for software development companies.

(b) Write a Java program to create a colour as a background.

---------- Cut here ----------

10. (a) Write Java program using Applet to create Forms and its Controls.

(b) Write a VBScript, which displays a Calender for the given Month and Year.

DE–9861

Sub. Code
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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2017.

UNIX AND SHELL PROGRAMMING – LAB

Time : Three Hours Maximum : 100 Marks

Examiner should select and give ONE question to each candidate by lot system.

1. (a) Write and execute a shell program to do the following operations :

   (i) Write the given string at given row, column in BOLD, normal and reverse.

   (ii) Search a given string using awk with various options.

   (b) Write a menu driven shell program to copy, edit, rename and delete a file.

---------- Cut here ----------

2. (a) Write a menu driven shell program for the following details :

   (i) Today's date and day

   (ii) User's system

   (iii) Quit from UNIX.

   (b) Write a shell program for six-function calculator by using YACC command.

3. (a) Write and execute a shell program to generate leave status report of an organization. Assume your own data.

(b) Write a Shell program to sum up the series

$$\frac{1}{1!}+\frac{3}{3!}+\frac{5}{5!}+\ldots+\frac{49}{49!}$$

---------- Cut here ----------

4. (a) Develop a Shell program using 3 arguments to take the pattern as well as input and output file names. If the pattern is found, display as "pattern found" else display as "Error message". Also check if right number of arguments are entered.

(b) Write a menu driven shell program to perform the following tasks :

(i) enter the sentence in a file

(ii) search a given whole word in an existing file

(iii) quit.

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5. (a) A hospital maintains patient details. Write a Shell program to perform the following operations :

(i) List the patient details, Blood group-wise

(ii) Patient's list agewise, between 20 and 50

(iii) Exit.

(b) Write a Shell program to find the sum of the series :

$$\frac{1}{1^2}+\frac{2}{2^2}+\ldots+\frac{10}{10^2}.$$

6. (a) Write and execute a Shell program to print monthly pay slips for employees in an organisation. Assume your own data and generate accordingly.

(b) Write a menu driven shell program to perform the following tasks :

(i) enter the sentence in a file

(ii) search a given whole word in an existing file

(iii) quit.

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7. (a) Write and execute a shell program to send mail to group of users by extracting their id's from /etc/group file.

(b) Write a shell program that accepts the name of a file from the standard input and then perform the following tests on it.

(i) enter 10 names in a file

(ii) sort the names

(iii) list unsorted and sorted names

(iv) quit.

8. (a) Write a Shell program which accepts the name of a file from the standard input and then perform the following tests on it :

   (i) file existence

   (ii) file readable

   (iii) file writable

   (iv) both readable and writable.

   (b) Write a Shell program which periodically monitors the disk for the existence of a file and then executes the program once the file has been located, using UNTIL statement.

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Sub. Code
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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated)
DEGREE EXAMINATION, DECEMBER 2017.

VISUAL PROGRAMMING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain in detail about Visual Basic Dialog Boxes.

2. Discuss various in-built and user defined functions in VB. Give example.

3. Discuss about the data types in Visual Basic with example.

4. Write a VC++ program to create simple calculator.

5. Explain about Object Linking and Embedding in VC++.

6. Write short note on : VC++ Serialization.

7. Discuss about MFC file handling methods.

8. How will you import VBX controls? Explain.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Write in detail about control arrays and dynamic arrays in Visual Basic with suitable example.

10. Discuss in detail about accessing database with data control using VB.

11. Discuss about any ten Visual Basic graphics creation commands with example.

12. Write in detail about how menu items connected to code with VC++ example.

13. Explain in detail about handling mouse and keyboard events in VC++.

14. Describe about the SDI and MDI environments in VC++.

15. Briefly explain about Dynamic data transfer functions and User Interface classes with examples.

DE–3044

Sub. Code
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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Year Integrated) DEGREE EXAMINATION, DECEMBER 2017.

SOFTWARE PROJECT MANAGEMENT AND METRICS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write about conventional software management performance.

2. Explain how does improving team effectiveness would help software economics.

3. Identify and explain the principles of modern software management.

4. Compare and contrast engineering artifacts with pragmatic artifacts.

5. Elucidate the technical perspective of model based software architectures.

6. Write about software process workflows.

7. Narrate the concept of process automation tool "Automation buildings blocks".

8. Identify the quality indicators that play vital roles in project control and instrumentation.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Give a detailed description on evolution of software economics.

10. List and explain the various life cycle phases in detail.

11. Describe the concept of artifact sets.

12. What are checkpoints of the process that would influence the software management framework? Explain.

13. Discuss the various responsibilities and project organizations that are considered as software management disciplines.

14. Give a detailed description on work breakdown structure.

15. What are process discriminants that tailors the process? Explain in detail.

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DE–3045

Sub. Code
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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2017.

WEB TECHNOLOGY

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. How does dynamic HTML is used in Javascript? Explain with sample code segment.

2. Write about XML processors usage in DOM and SAX.

3. Discuss the concept of Java Beans API.

4. What are major security issues that influence web servelet?

5. Give a note on JSP application design with MVC.

6. With sample code, discuss the process of error handling and debugging shared data between JSP pages.

7. Compare and contrast sessions with cookies related to web services.

8. How do databases are accessed in JDBC? Explain in detail.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Give a detailed description on HTML tags and cascading style sheets with example codes.

10. Write notes on the following concepts :

    (a) XML schemas. (8)

    (b) DTD features used in XML. (7)

11. (a) List down the advantages of Javabeans. (6)

    (b) Life cycle of web servelet. (9)

12. Explain how JSP application development pages are made with dynamic content using scripting elements.

13. Narrate the procedure of reading servelet parameters with initialization parameters, through an example code.

14. Elaborate the concepts of various conditional processing followed in JSP.

15. Explain in detail about the database actions and deploying Java Beans made in a JSP page.

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DE–3046

Sub. Code
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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2017.

DATA WAREHOUSING AND MINING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the procedure of implementing data warehouse.

2. Discuss the concept of data reduction.

3. Give a note in KDD process.

4. Narrate the concept of support vector machines.

5. Write notes on grid based methods.

6. Discuss the various prediction methods.

7. Compare and contract complete link with average link method.

8. List and explain the various applications of web mining.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Write notes on the following :
   (a) Classification of data mining systems. (7)
   (b) Major issues in data mining. (8)

10. Narrate the procedure of "Data reduction" used as data preprocessing.

11. Discuss the various association rules followed in large data mining.

12. Give a neat description on clustering the high dimensional data method.

13. Elucidate the process of classification by back propagation.

14. Write notes on the following algorithms related to clustering :

    (a) K-means (7)

    (b) PAM (4)

    (c) Genetic algorithm. (4)

15. Explain the concepts of web usage mining research and ontology based web mining research.

DE–3047

Sub. Code
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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2017.

DISTRIBUTED COMPUTING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Describe the architectural model employed in distributed system.

2. Discuss the issues related to datagram communication.

3. What are the main characteristics of distributed event-based system? Explain.

4. Write about firewalls.

5. Write the procedure for creation of a new process.

6. Describe the requirements and pitfalls in the design of distributed services.

7. Write a note on Domain Name System.

8. Illustrate the problems associated with aborting transactions.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Describe the challenges to be faced at the time of designing distributed system.

10. Explain the role of each component in RMI.

11. Describe the design of scret-key and public-key algorithm.

12. Explain the Cristian's method for synchronizing clocks.

13. Explain the architecture of X.500 directory service.

14. Write about concurrency control in distributed transactions.

15. Discuss about group communication in detail.

DE–9862

Sub. Code
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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2017.

VISUAL BASIC AND VC++ – LAB

Time : Three hours Maximum : 100 marks

Each student has to select ONE question by lot system.

1. (a) Using Flex Grid control display the multiplication and addition table of 20 rows and columns.

   (b) Write a Visual C++ program to find out whether a mouse is attached or not; and if attached, how many buttons are present in it.

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2. (a) Using built-in ActiveX control, develop the windows application like NOTEPAD with file and edit menus and also display the floating menu whenever necessary.

   (b) Write a Visual C++ program to create a list box in window.

3. (a) Using ActiveX DLL or EXE add a class module that would perform the following functions:

   (i) Test whether the given number is perfect or not

   (ii) Test whether the given number is Armstrong or not

   (iii) Find the factorial of the given number.

   (iv) Find the sum of digits in a given number.

   (b) Write a Visual C++ program to generate a status bar and show the status of Caps Lock, Num Lock, and Scroll Lock in it.

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4. (a) Create a VB application with a Drive List Box, Directory List Box and File List Box.

   (i) Select any EXE file & execute it

   (ii) Select any picture file and load it into an image box.

   (iii) Filter the file in the file list box with the filter pattern chosen from a combo box.

   (b) Write a Visual C++ program using MFC to get the status of the shift and toggle keys.

5. (a) Using ActiveX control create a Text Box that accepts only numeric value and has the following properties:

   (i) Background of the textbox.

   (ii) Foreground of the textbox.

   (iii) Text property of the textbox.

   (iv) Resize the textbox at the standard application without using properties.

   (b) Write a Visual C++ program to fill background of the client area with a bitmap.

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6. (a) Create a DHTML application.

   (b) Write a Visual C++ program using MFC to handle Windows messages.

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7. (a) Using a control array, create a simple calculator capable of doing the following operations:

   (i) Addition

   (ii) Subtraction

   (iii) Multiplication

   (iv) Division

   (v) Square

   (vi) Square root

   (vii) Modulus.

   (b) Write a Visual C++ program to create a list box in window.

8. (a) Create a pay table in employee database with Empno, Empname, Basicpay, HRA, DA, PF, LIC, GP and NP fields. Use the following formulae for computations.

HRA = 10% of BP

DA = 5% of BP

PF = 3% of BP

LIC = 5% of BP

GP = BP+DA+HRA

NP = GP–(PF+LIC)

Using Remote Data Access Object, Implement the following operations:

(i) Insert a record

(ii) Search and delete a record.

(iii) Modify the record.

(iv) Display all the employee records whose names are starting with the letter "M".

(b) Write a Visual C++ program to create a window of desired size using MFC.

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DE–9863

Sub. Code
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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2017.

WEB TECHNOLOGY – LAB

Time : Three hours Maximum : 100 marks

Each Student has to select ONE question by lot system.

1. (a) Create a HTML table with rows and columns and split them using Rowspan and Colspan.

   (b) Write an ASP program to prepare employee pay bill using Java script.

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2. (a) Create a web page for the first page of a news paper. Provide hyper links to other pages. Align the text and format with different colors.

   (b) Write an ASP program to prepare student performance evaluation document using Java script.

---------- Cut here ----------

3. (a) Write XML document to display your bio-data. Write XSL style sheet and attach it to the XML document. Validate the document using DTD or XSD.

   (b) Write a Servlet program to display “Hello World”.

4. (a) Create a HTML table with rows and columns and split them using Rowspan and Colspan.

   (b) Create an Image animation using Java script.

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5. (a) Create a web page for the first page of a news paper. Provide hyper links to other pages. Align the text and format with different colors.

   (b) Write a JSP program to display student mark list using JDBC.

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6. (a) Write an XML document to display your bio-data. Write an XSL style sheet and attach it to the XML document. Validate the document using DTD or XSD.

   (b) Develop an Application for JSP – Servlet communication.

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7. (a) Create a web page for the first page of a news paper. Provide hyper links to other pages. Align the text and format with different colors.

   (b) Write Servlet program to send Email message.

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8. (a) Write an ASP program to prepare student performance evaluation document using Java script.

   (b) Write an ASP program to prepare employee pay bill using Java script.

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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2017.

SOFTWARE TESTING AND REUSE

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What are the five Phases of software testing? Explain.

2. Explain the three kinds of loops.

3. What is data flow testing? Explain the motivation and assumptions of data flow testing.

4. Describe the model of domain testing.

5. Explain any eight rules of path product.

6. What is regular expression? Explain.

7. Explain briefly about state graph with an example.

8. Write a short note on JMeter and Winrunner tools.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Define the term "Bugs". What are the types of Bugs? Explain.

10. What is path instrumentation? Discuss.

11. Explain the Transaction Flow Testing Techniques.

12. Explain about domain and interface testing.

13. What is Karnaugh-Veitch chart? Explain with examples.

14. Write the node-reduction algorithm and explain it.

15. Discuss about the performance testing of a database application.

DE–3049

Sub. Code
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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2017.

CRYPTOGRAPHY AND NETWORK SECURITY

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. Explain the Security Attacks.

2. Write about the approaches to Message Authentication.

3. Discuss the conventional encryption principles.

4. Explain the RSA public-key encryption algorithm.

5. Explain briefly about the X.509 directory authentication service.

6. Discuss the principal services provided by PGP.

7. Write about the nature and the types of viruses.

8. Write about the types of firewalls.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Write in detail about security services.

10. Explain the various cipher block modes of operation.

11. Discuss in detail about secure Hash functions.

12. Explain briefly about Kerberos.

13. Write briefly about secure/multipurpose Internet Mail Extension (S/MIME).

14. Write briefly about secure socket layer.

15. Explain in detail about intrusion detection systems.

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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2017.

SOFTWARE QUALITY ASSURANCE AND STANDARDS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. Describe the relationship between different quality measures.

2. Depict the process involved in Goal Question Metric (GQM) approach.

3. Enumerate the typical goals and objectives of quality engineering.

4. Illustrate the relationship between different quality tasks.

5. Describe various types of quality tools.

6. Explain how the reliability growth model is used for quality assessment.

7. Write a note on 'Customer Satisfaction Analysis'.

8. Compare and contrast CMM and CMMI.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Elaborate different views of software quality.

10. Explain two principal types of hierarchical model to compare software quality in different situations.

11. Discuss the steps involved in software quality assurance plan.

12. What is the use of CASE tools? Describe the function of various CASE tools.

13. Describe the method of preventing, discovering and removing defects in software project with illustration.

14. Elaborate the use of Rayleigh model framework for quality management.

15. What six sigma? Elaborate the method of using six sigma concepts in maintaining quality standards.

DE–3051

Sub. Code
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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2017.

MOBILE COMMUNICATIONS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. State and explain the basics of Satellite Systems.

2. Highlight the advantages of Digital Video Broad Casting System.

3. Define 802.11 standard and highlight its advantages.

4. Write short notes on Bluetooth Technology.

5. Briefly explain Ad-hoc Network.

6. Explain about IPV6.

7. What is Timeout Freezing? Explain briefly.

8. With a neat sketch, explain the Architecture of Datagram Protocol.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss on the following :
   (a) SDMA
   (b) FDMA
   (c) TDMA.

10. Describe the Wireless ATM Working Group with Services.

11. Discuss on Tunneling and Reverse Tunneling.

12. Write notes on :

    (a) Snooping TCP

    (b) Mobile TCP

    (c) Indirect TCP.

13. Discuss in detail the Wireless Telephony System.

14. With examples, explain :

    (a) Digital Video Broadcasting

    (b) Mobile IP Goals.

15. Give a brief account on :

    (a) Transaction Oriented TCP.

    (b) Transport Layer Security.

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Sub. Code
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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2017.

OPEN SOURCE ARCHITECTURE

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. What are the advantages and disadvantages of Open Source Soft wares?

2. How do you grant privileges for assigning permissions to files and folders in Linux?

3. How do you set up email server? Explain.

4. What are the utilities available in Linux to manage day by day tasks?

5. How to set up a firewall for an enterprise?

6. What is the use of autoconf and autogen? Discuss.

7. Give a brief note on X windows server architecture.

8. How do you generate statically linked libraries? Discuss.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the different modes of connecting internet.

10. How do you configure disk partitions and file system in Linux? Explain.

11. Explain the process of setting up proxy services and printers services.

12. Explain the configuration and utilities in X window system.

13. What is source code versioning? How do you handle source code versioning?

14. Briefly discuss on graphical debuggers and memory debugging.

15. Discuss the features of python programming.

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DISTANCE EDUCATION

M.Sc. Software Engineering (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2017.

OPEN SOURCE LAB

Time : Three hours Maximum : 100 marks

Examiner should select and give one question to each candidate by lot system.

1. (a) Write a PHP program using classes to create a table.

   (b) Write a PHP program to upload a file to the server.

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2. (a) Write a server side PHP program that displays marks, total, grade of a student in tabular format by accepting user inputs for name, number and marks from a HTML form.

   (b) Write a PHP program that adds products that are selected from a web page to a shopping cart.

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3. (a) Write a PHP program interface to create a database and to insert a table into it.

   (b) Write a PHP program to access the data stored in a mysql table.

4. (a) Write a PHP program to create a directory and to read contents from directory.

   (b) Write a shell program to find the details of an user session.

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5. (a) Write a Shell program to change the extension of a given file.

   (b) Create a mysql table and execute queries to read, add, remove and modify a record from that table.

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6. (a) Write a PHP program to access the data stored in a mysql table.

   (b) Write a PHP program to determine the given year is a leap year or not.

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7. (a) Write a shell program to find the details of an user session.

   (b) Write a PHP program using classes to create a table.

---------- Cut here ----------

8. (a) Write a shell program to change the extension of a given file.

   (b) Write a PHP program to access the data stored in a mysql table.

DE–9865

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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2017.

CASE TOOLS LAB

Time : Three hours Maximum : 100 marks

One question should be given to each candidate by lot system.

Each sub-division carries 50 marks.

1. (a) Design build and test class diagrams for a software project.

   (b) Design the stock maintenance system.

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2. (a) Design a model for identification of deliverables.

   (b) Design the Real-Time scheduler.

---------- Cut here ----------

3. (a) Design a model for specification of project objectives and infrastructure.

   (b) Design the Quiz system.

---------- Cut here ----------

4. (a) Design a model for identification of project scope.

   (b) Design the Payroll system.

5. (a) Develop a specification model for modules of a projects.

   (b) Design the Student Mark Analyzing System.

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6. (a) Design use diagrams and activity diagrams for software projects.

   (b) Design the Quiz system.

---------- Cut here ----------

7. (a) Design a model for software development and debugging.

   (b) Design the Online Ticket Reservation System.

---------- Cut here ----------

8. (a) Design data modeling use work products.

   (b) Design the ATM system.

---------- Cut here ----------

9. (a) Design a model for software requirement analysis.

   (b) Design the Quiz system.

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10. (a) Perform program analysis and project planning.

    (b) Design the stock maintenance.